

MAHMOUD ELLOUMI

Computer Engineering Student • Generative AI & AI Agents

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PROFILE

Computer engineering and applied mathematics student at ENIS, I design **full-stack** applications that integrate **generative AI**, **AI agents** and computer vision. Through two internships and team projects, I have integrated LLMs (Ollama, Mistral AI) and n8n workflows to turn concrete business needs into intelligent solutions.

EDUCATION

Engineering Cycle in Computer Science and Applied Mathematics | *National Engineering School of Sfax (ENIS)* **2024 – Present**

Preparatory Cycle in Technology • top 100 in the national ranking | *IPEIS, Sfax* **2022 – 2024**

Baccalaureate in Technical Sciences | *15 November 1955 High School, Sfax* **2022**

PROFESSIONAL EXPERIENCE

Full-Stack & AI Developer Intern | IDSOFT (ID Software Solutions), Sfax **07/2026 – 08/2026**

AI-enhanced after-sales service management app • React, Spring Boot, PostgreSQL, Ollama

- Built an after-sales web app (7-state service-request lifecycle) for 3 roles: customer, technician, manager.
- Integrated 6 local generative AI features (Ollama, no data sent to third parties): troubleshooting assistant, natural-language data assistant, request classification, best-technician suggestion, report-writing assistance.
- Designed a layered Spring Boot REST API (Swagger), secured with JWT, TOTP, BCrypt and role-based access control.

Mobile Developer Intern | LOGICOM Informatique, Sfax **07/2025 – 08/2025**

Product scanning mobile app • Flutter

- Built a Flutter mobile app that scans a product's QR code to display its available prices and product details.
- Connected the app to the company database to retrieve prices and records for 40 products.

PROJECTS

SPARKY – Intelligent Business Document Processing (PFA) **05/2026**

Team of 3, Scrum (9 sprints) • Angular, FastAPI, Mistral AI, PostgreSQL, n8n

- Designed the n8n workflow (11 nodes, Google Sheets, email) and the Angular 20 frontend: scanner, price comparison, dashboard.
- Extracted invoices, quotes and purchase orders with Mistral AI in 3–8 s each, vs 120–180 s for the n8n + Ollama prototype.
- Compared 3 suppliers across 10 items in under 1 minute, vs about 20 minutes by hand.

TRAFFICEYE – Intelligent Traffic Monitoring **04/2025**

Course project • Python, YOLO, OpenCV

- Detected and classified vehicles in real time on video streams with YOLO and OpenCV.
- Built a traffic analysis dashboard: vehicle counts and breakdown by vehicle type.

TERRANAUT – Drone-Based Detection of Sick Animals **04/2024**

Course project • Computer vision, Deep Learning, thermal imaging

- Spotted sick animals from thermal images captured by drones.
- Applied computer vision and deep learning (object detection) to thermal image analysis.

TECHNICAL SKILLS

AI & LLMs: LLMs (Ollama, Mistral AI, Qwen), Generative AI, AI agents, Machine Learning (scikit-learn)
Vision & Automation: Deep Learning, OpenCV, YOLO, PyTorch, OCR, n8n
Languages: Python, Java, JavaScript/TypeScript, C
Web & Mobile: React, Angular, Spring Boot, FastAPI, Node.js, Flutter
Data & DevOps: PostgreSQL, MySQL, MongoDB, Kafka, Power BI, Docker, Kubernetes, Git, Linux, Scrum

CERTIFICATIONS

NVIDIA Deep Learning • Cisco C Advanced • Scrum (SFC) • Hedera (HCF) • Microsoft MS-900

VOLUNTEERING (IEEE)

Photographer – IEEE ENIS SB (2026) • **Treasurer & Webmaster** – IEEE Cyber Security ENIS (2025)
Social Media Manager & Photographer – TSYP 13 • **Organizer** – Pitch Yourself for a Career

LANGUAGES & INTERESTS

Arabic: native • **French:** upper-intermediate • **English:** intermediate • **Spanish:** elementary

Interests: photography, art, sports, innovation